

TAMIL NADU STATE BOARD - STANDARD 11 MATHEMATICS VOLUME 1

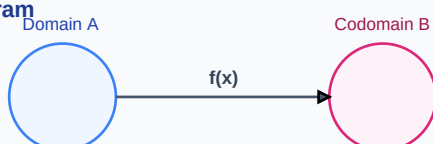
CHAPTER 5: BINOMIAL THEOREM, SEQUENCES AND SERIES

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Binomial Theorem, Sequences and Series

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Binomial Expansion	An algebraic expansion of powers of a binomial $(x + a)^n$ using combinations coefficients.
Arithmetic Progression (AP)	A sequence where the difference between consecutive terms is a constant d .
Geometric Progression (GP)	A sequence where the ratio of consecutive terms is a constant non-zero common ratio r .

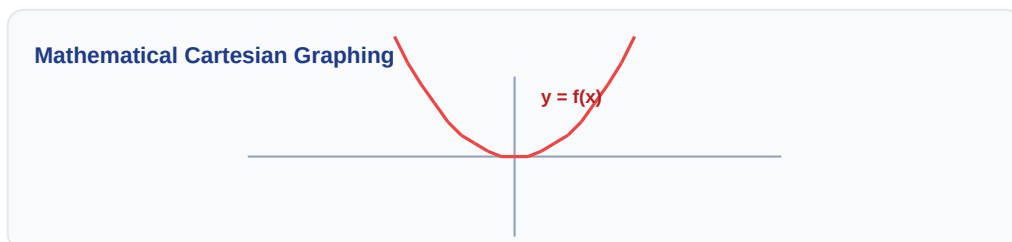
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Binomial Theorem: $(x+a)^n = \sum_{r=0}^n {}^nC_r \cdot x^{n-r} \cdot a^r$
- General Term of AP: $t_n = a + (n-1)d$ | Sum: $S_n = \frac{n}{2}[2a + (n-1)d]$
- General Term of GP: $t_n = a \cdot r^{n-1}$ | Sum: $S_n = \frac{a(r^n - 1)}{(r-1)}$
- Infinite GP Sum: $S = \frac{a}{1-r}$ for $|r| < 1$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the middle term in the expansion of $(x + 2/x)^8$.

Step-by-step Solution:

Number of terms is 9, so middle term is t_5 .

$$t_5 = {}^8C_4 * x^{(8-4)} * (2/x)^4 = {}^8C_4 * x^4 * 16/x^4 = 70 * 16 = 1120.$$

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: If the sum of a series is $S_n = 2n^2 + 5n$, find the 10th term and show the sequence is an AP.

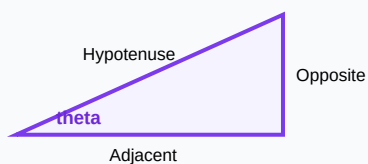
Analytical Solution Strategy:

$$t_n = S_n - S_{(n-1)} = (2n^2 + 5n) - (2(n-1)^2 + 5(n-1)) = 4n + 3.$$

$$\text{Common difference: } t_n - t_{(n-1)} = (4n+3) - (4(n-1)+3) = 4 \text{ (constant, hence AP).}$$

$$10\text{th term: } t_{10} = 4(10) + 3 = 43.$$

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Write the binomial expansion of $(2x - 3y)^4$.

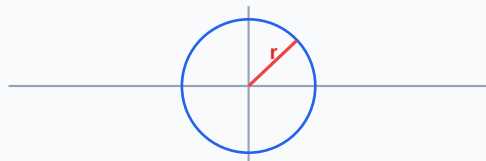
Q2 (2-Mark): Find the 15th term of the AP: 5, 8, 11, 14, ...

Q3 (2-Mark): Find the sum of the infinite geometric series: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$

Q4 (5-Mark): Define Arithmetic Mean (AM) and Geometric Mean (GM).

Q5 (5-Mark): Insert three geometric means between 2 and 32.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Financial Calculations

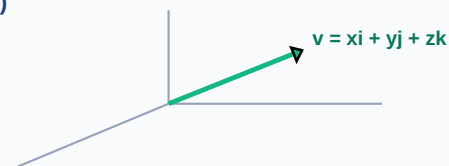
Compound interest values are derived using binomial series approximations for daily compounding rates.

Application Case: Acoustic Scales

Musical octaves are geometric progressions of frequencies, doubling pitch levels per octave.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

